

A PROJECT TITLE

ON

DDoS Attack Detection Using LSTM and Autoencoder-Based Anomaly Detection

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

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by

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Name of Title: DDoS Attack Detection Using LSTM and Autoencoder-Based Anomaly Detection

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1. Abstract:

Distributed Denial-of-Service (DDoS) attacks continue to be one of the most disruptive threats to network availability, overwhelming servers with high-volume malicious traffic and causing significant financial and operational damage to organizations. Traditional signature-based intrusion detection systems struggle to identify novel or evolving attack patterns, creating a need for intelligent, data-driven detection mechanisms. This project proposes a hybrid deep learning approach that combines Long Short-Term Memory (LSTM) networks with an Autoencoder-based anomaly detection framework to identify DDoS attacks in network traffic. The LSTM component captures temporal dependencies and sequential patterns within traffic flows, while the Autoencoder learns a compressed representation of normal traffic behaviour and flags deviations as potential anomalies based on reconstruction error. By fusing sequence modelling with unsupervised anomaly detection, the system is able to detect both known and previously unseen attack variants without relying solely on labelled attack signatures. The proposed model is trained and evaluated on the CICDDoS2019 dataset, a widely used benchmark containing labelled records of normal traffic and multiple categories of DDoS traffic such as SYN flood, UDP flood, and reflection-based attacks. Data preprocessing steps including feature scaling, normalization, and sequence windowing are applied before feeding the data into the network. The performance of the proposed system is assessed using accuracy, precision, recall, F1-score, and ROC-AUC as evaluation metrics, providing a comprehensive measure of the model's detection capability under the class-imbalanced conditions typical of real-world network traffic.

2. INTRODUCTION:

The rapid expansion of internet-connected devices and cloud-based services has significantly increased the attack surface for Distributed Denial-of-Service (DDoS) attacks, which remain among the most severe threats to the availability and reliability of modern networks. A successful DDoS attack can render critical services inaccessible, resulting in substantial revenue loss, reputational damage, and disruption of essential operations for businesses and institutions. Motivated by the growing sophistication and volume of such attacks, and by the limitations of conventional rule-based detection systems in adapting to new and evolving attack strategies, this project explores the use of deep learning techniques to build a more robust, adaptive, and reliable DDoS detection system.

Machine learning, and more specifically deep learning, has emerged as a powerful tool for network intrusion detection due to its ability to automatically learn discriminative features from raw or minimally processed traffic data. Long Short-Term Memory (LSTM) networks, a class of recurrent neural networks, are particularly well suited to modelling the temporal and sequential nature of network traffic, allowing the detection of attack patterns that unfold over time. Complementing this, Autoencoder-based anomaly detection provides an unsupervised mechanism to identify traffic that deviates from learned normal behaviour, making it effective at flagging previously unseen or zero-day attack variants. This project integrates both approaches into a unified pipeline that first uses the LSTM to model traffic sequences and then applies the Autoencoder to detect anomalous reconstruction patterns, enabling accurate and adaptive detection of DDoS attacks.

3. SOFTWARE REQUIREMENTS:

The proposed DDoS detection system is implemented using Python-based deep learning and data-processing libraries within a standard development environment. Model development and experimentation are carried out using TensorFlow/Keras for building the LSTM and Autoencoder architectures, along with supporting libraries such as NumPy, Pandas, and Scikit-learn for data preprocessing, feature scaling, and performance

evaluation. Given the computational demands of training deep learning models on large-scale network traffic datasets, a system equipped with a dedicated GPU is recommended in order to reduce training time and enable efficient experimentation. The specific software and hardware requirements used for this project are summarized in the table below.

Category	Requirement
Programming Language	Python 3.x
Deep Learning Framework	TensorFlow / Keras
Supporting Libraries	NumPy, Pandas, Scikit-learn
Dataset	CICDDoS2019
Hardware	System with a dedicated GPU (recommended)
Development Environment	Jupyter Notebook / Python IDE

Signature of the Guide

Course Instructor